RESEARCH / PROJECT PROPOSAL

Internship Navigator: A Workflow-Based Internship Management System

Project title	Internship Navigator: A Workflow-Based Internship Management System
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Stakeholder / partner	Internship Coordinator and Academic Advisors of the program (interview access to be confirmed)
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Keywords	internship management; requirements traceability; acceptance criteria; audit trail; role-based access

Executive Summary

Internship Navigator is a role-based web application designed to simplify and organize the university internship process, from host-company proposals to final completion. Currently, many steps are handled manually across different communication channels, while the existing system mainly records internship information, daily tasks, and host-company evaluations. It does not support the entire approval, document management, supervision, and completion process.

The proposed system will provide a centralized view of student status, required documents, next steps, and supervision records. It will include a **Readiness Gate** to ensure all requirements are met before an internship begins and a **Completion Gate** to ensure all conditions are satisfied before final closure.

Development will begin with stakeholder consultation and a prototype focusing on status tracking, host-company approval, readiness checks, and decision history. Subsequent stages will expand the system with the core internship workflow, map-supported supervision, offline synchronization, reporting, and deployment, subject to feasibility and stakeholder approval.

3-minute pitch

Internship coordination involves four different roles, but there is no single place to track the entire process. Staff often need to check messages, documents, and spreadsheets to understand each student's progress. This can make the next steps unclear and decisions difficult to trace.

Our project, Internship Navigator, is a workflow-based web system that connects those steps without replacing the existing system used for daily task logging. It will provide role-specific dashboards, a semester-wide status view, document tracking, and a clear history of decisions.

Two key features are the **Readiness Gate**, which ensures that placement approval, required documents, training eligibility, and instructor assignment are complete before the internship begins, and the **Completion Gate**, which verifies all completion requirements before the Coordinator confirms final closure.

For this seminar, we will not claim to deliver the whole platform. We will deliver a thin working prototype, a validated workflow and state model, test cases linked to requirements, and an evaluation using realistic scenarios with synthetic data. We are seeking approval of this staged scope and access to relevant stakeholders for workflow validation and prototype evaluation.

1. Title, Team & Advisor

Working title: Internship Navigator: A Workflow-Based Internship Management System

One-sentence pitch: Internship Navigator is a workflow-based internship management system that streamlines the entire internship process—from student readiness and document approval to supervision and final completion—through one centralized platform.

Roles: Lead/Requirements owns the requirement baseline, assumption register and traceability matrix; Design & Prototype owns the state model, architecture and prototype slice; Testing/Evaluation owns acceptance tests and the reviewer protocol. Faculty advisor: [Aj. Prasara Jakkaew] — checkpoint decisions (Go / Go-with-revisions / Rework) and SE rigor gate.

2. Problem, Context, and Evidence

2.1 Background and problem statement

The internship journey includes company proposal and presentation, training eligibility, three required documents, instructor assignment, a three-month placement, supervision, problem-case handling, report review, and final closure. The SRS states that the existing core internship system does not manage the entire journey. As a result, staff use manual follow-up and multiple communication channels to assemble the current status.

- Status visibility: students may not know their current stage or the next action they must take.
- Document and readiness tracking: staff must check training evidence and three required letters before the internship begins.
- Supervision planning: company locations and assignments need to be organized to support online or on-site supervision.
- Decision traceability: failed presentations, appeals, report revisions, problem cases, and instructor changes need history rather than silent overwriting.
- Completion control: staff must verify several conditions and manually confirm final closure.

2.2 Evidence currently available

The proposal must distinguish evidence already present in the project materials from evidence still to be collected. The current baseline is preliminary and should not be presented as a completed quantitative study.

Evidence source	What it currently supports	Limitation / next action
Stakeholder feedback recorded in SRS	Staff report that the full process is not managed in one website and many steps are handled manually.	Record interview date, role, consent, and notes in the research log; confirm the wording with the stakeholder.
Current-system boundary described in SRS	The existing system focuses on internship information/daily tasks and host-company evaluation; broader workflow remains outside it.	Validate the boundary with the Coordinator and confirm any existing spreadsheets or tools.
Requirements change history in proposal draft	The project requirements have changed during specification work, showing a need for a stable baseline and traceability.	Maintain a dated change log; do not treat requirement growth alone as proof of operational impact.
Open questions in SRS	Route automation, reminder timing, host-company evaluation ownership, file storage, and quality targets need confirmation.	Use structured stakeholder interviews and record decisions, owners, and due dates.

3. Related Work and Gap Analysis

3.1 Internship management systems

The proposal's initial literature map describes prior university and internship systems that support functions such as documentation, communication, placement allocation, monitoring, and reporting. These systems show that web-based workflow support can reduce scattered information and repeated manual work. However, the abstracts summarized in the current reading set do not establish that all of them provide explicit readiness and completion gates, immutable decision history, and a requirement-to-test chain.

3.2 Requirements traceability

Requirements traceability links a requirement to its origin and to evidence that it has been implemented and tested. Reviews discuss the benefits and challenges of maintaining these links, including accuracy and effort. For this student project, a lightweight, human-maintained link from functional requirement to acceptance criterion and test case is more practical than introducing a complex automated traceability-recovery tool.

3.3 Gap analysis

Area	Common focus in reviewed work	Gap this project addresses
Internship workflow	Documents, communication, allocation, monitoring, and assessment.	A connected journey with explicit Readiness and Completion Gates.
Decision history	System-specific records and reports.	A consistent audit trail preserving actor, time, reason, and previous/new values.
Requirements quality	Traceability methods often focus on recovering links at scale.	A manageable FR → acceptance criterion → test chain maintained as requirements change.
Workflow evaluation	Usability or system effectiveness may be reported.	Scenario-based evaluation of status clarity, gate correctness, role fit, and auditability.

3.4 Project gap and contribution

The project contribution is not simply another internship portal. It is a staged, testable workflow model that makes blockers visible, prevents invalid status transitions, preserves decision history, and evaluates whether the resulting status view supports staff and students. This gap is preliminary and will be refined after the planned stakeholder interviews and a fuller review of the cited studies.

4. Objectives and Research Questions

4.1 Objectives

- Validate and baseline the internship workflow, role responsibilities, business rules, and open assumptions using stakeholder evidence.
- Design an explicit status lifecycle and gate rules for readiness and completion, including audit requirements and exceptional-case handling.
- Build a thin prototype for status tracking, host-company proposal/presentation decisions, readiness evaluation, and audit history.
- Evaluate the prototype through acceptance tests and scenario walkthroughs, then document findings and a feasible SP1/SP2 implementation roadmap.

4.2 Research / project questions

RQ1: How can Internship Navigator represent the internship journey as explicit, auditable states and gate-controlled transitions across four user roles?

RQ2: To what extent does the prototype help reviewers identify each student's current stage, blockers, and next action while preserving decision traceability?

4.3 Expected contribution

- A stakeholder-reviewed workflow and state model.
- A traceable requirement, acceptance-criteria, and test baseline.
- A working thin prototype with synthetic data.
- Evaluation findings, limitations, and a staged roadmap for Senior Project 1 and 2.

5. Scope and Boundaries

5.1 Seminar-term scope

- Role-aware login concept and dashboards for the four roles, using test accounts.
- Student status page and semester-wide status table with status filters.
- Host-company proposal submission, presentation scheduling, advisor pass/fail decision, and visible decision history.
- Readiness Gate logic showing each of its four checks and why a student is blocked.
- Audit history for status changes and decisions in the implemented slice.
- Synthetic test data, acceptance tests, scenario walkthroughs, and documentation.

5.2 Full-system target scope across SP1/SP2

- Training-session tracking and certificate verification; tracking the Request Letter, Acceptance Letter, and Referral/Placement Letter.
- Company location pins, route-group support, and Coordinator assignment of one Supervising Instructor per company/route group.
- Supervision scheduling, scoring, interview records, problem cases, and offline draft synchronization.
- Student declaration that daily task data was entered into the existing core system; no direct integration in the current SRS.
- Internship end, host-company evaluation, report submission/revisions, leave/evidence checks, and Completion Gate.
- Notifications, organization records, and other Should-have features according to priority and stakeholder confirmation.

5.3 Out of scope / boundaries

- Replacing the existing core internship system or integrating with it through an API.
- Host-company self-service accounts unless stakeholders change the current assumption.
- Leave approval; the system records leave evidence after the student handles leave with the host company.
- Production deployment or use as an official decision-making record during the seminar prototype.
- Digital signatures and compensation/benefit processing.

5.4 Key assumptions and dependencies

The university will provide current internship rules, official document templates, scoring rubrics, and letter formats. Google Maps access and any applicable API configuration are dependencies for later map functionality. The planned internship duration is three months. The SRS default of five training sessions is configurable, and reminder timing, minimum supervision count, notification channel, file storage, and route-automation level require stakeholder confirmation.

6. Method, Design, Technical Plan and Diagrams

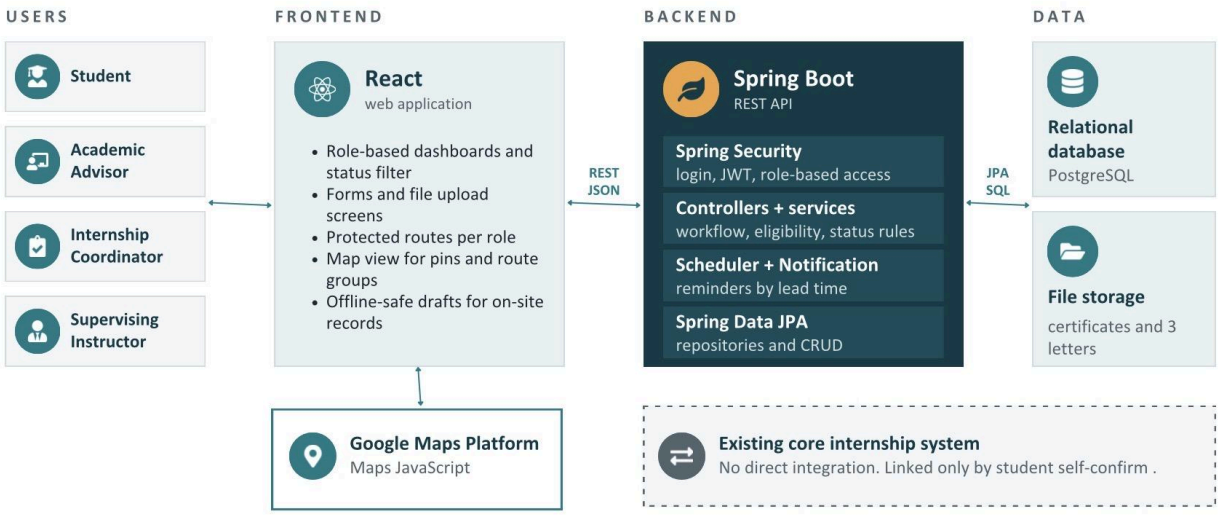
6.1 Method

Step	Activity	Deliverable
1. Discover	Review SRS and proposal; interview stakeholders; log assumptions and evidence.	Evidence summary and assumption register.
2. Specify	Agree roles, status names, gate conditions, exceptional cases, and acceptance criteria.	Reviewed workflow and status model.
3. Design	Define API boundaries, data model, access control, audit events, and prototype screens.	Architecture and design specification.
4. Build	Implement the thin slice with synthetic records and role-based permissions.	Runnable prototype.
5. Verify	Run automated acceptance tests and negative/edge-case scenarios.	Test report linked to requirements.
6. Evaluate	Run 3–4 scenarios per reviewer; collect 1–5 ratings and qualitative feedback.	Evaluation results and limitations.
7. Handoff	Update the SRS, traceability matrix, risk register, and roadmap.	Seminar defense package and SP1/SP2 backlog.

6.2 Proposed system architecture

Approach Overview: This project uses the existing Internship Management System documentation as the foundation for defining and developing a focused, testable prototype. The seminar will focus on a structured workflow model that tracks student status, defines clear readiness and completion conditions, and records decisions with a complete history. The prototype will be evaluated through stakeholder feedback and systematic testing, supported by traceability from requirements to acceptance criteria and test cases. Broader features, including system integrations, on-site supervision tools, and automated letter generation, are reserved for future development. The approach emphasizes transparency, accountability, and verifiable workflow rules rather than relying on implicit logic embedded in the code.

The system will use React for the frontend, Spring Boot REST API for backend workflow and security, Spring Security/JWT for authentication and role-based access, Spring Data JPA for data access, and PostgreSQL for structured records. Google Maps Platform will support company pins and route grouping. On-site supervision drafts will be stored locally and synchronized when connectivity returns.



6.2.1 Input/output contracts

Artifact	**Key fields / content**
Student record	student_id; user_id; student_number; program; year_level; internship_status
Proposal record	proposal_id; student_id; company_id; presentation_datetime; result; comments; evaluated_by; evaluated_at
Document record	document_id; student_id; document_type; file_url; version_number; submission_status; validation_status
Training record	training_record_id; student_id; session_name; certificate_file_url; verification_status; verified_by
Readiness Gate output	gate_result {READY | NOT_YET_COMPLETE | URGENT}; condition checks {proposal_approved, documents_complete, training_passed, instructor_assigned}; blocking_reasons[]
Audit event	event_id; actor_id; role; timestamp; action; reason; old_value; new_value; gate_conditions_snapshot; result

Dashboard view	student status; semester-wide table; filters; urgent flags; next action; blocker summary
Report package	status export; decision history; gate condition snapshot; export timestamp

6.2.2 Technology choices — and why

Layer	**Choice**	**Why (fit, not fashion)**
Frontend	React + charting library	Role-based dashboards, status tables, and export without overbuilding a platform
Backend	Spring Boot REST API	Workflow logic, role-based security, and transaction management
Auth / access	Spring Security + JWT	Role-based access enforced on the backend, not only in the UI
Data access	Spring Data JPA	Structured records, relationships, and audit history
Database	PostgreSQL	Reliable structured storage for status, decisions, and audit events
Maps (parked)	Google Maps Platform	Company pins and route grouping in SP1/SP2; modular so it can be stubbed this term
Offline sync (parked)	Local drafts + sync status	SP1/SP2; not required for the seminar thin slice

These choices optimize for a one-semester evaluable prototype. They deliberately avoid requiring production data, real student records, or microservices complexity inappropriate to the seminar scope.

6.2.3 Example gate rules

Rules are transparent, auditable, and stakeholder-linked — not black-box automation. Each rule returns a gate result, supporting conditions, and confidence. Examples below illustrate the format; final thresholds and conditions will be confirmed with the Coordinator and Advisor.

Rule G1 — Readiness Gate: IF proposal approved AND all three required documents present and verified AND training eligibility passed AND Supervising Instructor assigned THEN gate_result = READY; confidence = high; else gate_result = NOT_YET_COMPLETE with blocking_reasons[]; if the internship start date is within the configured urgent window and any required condition is incomplete, THEN display URGENT (a prioritization label, not a pass).The urgent window defaults to 14 calendar days before the planned start date and is configurable by the Internship Coordinator.

Rule G2 — Proposal rejection loop: IF advisor records result = REJECTED THEN status = MUST_FIND_NEW_PLACEMENT; previous proposal and decision remain linked; student may submit another proposal and schedule another presentation.

Rule G3 — Company cancellation after approval: IF host company cancels acceptance after approval but before start THEN status = MUST_FIND_NEW_PLACEMENT; original history remains linked; readiness gate resets to NOT_YET_COMPLETE.

Rule G4 — Report revision: IF report review result = REVISION_REQUIRED THEN earlier version remains available; completion gate stays blocked until report is passed.

Rule G5 — Completion Gate (design artifact this term): IF planned duration complete AND minimum supervision count satisfied AND host-company evaluation complete AND report passed AND leave records verified (if any) THEN gate_result = READY_FOR_CLOSURE; Coordinator must manually confirm final closure; gate result must never be silently rewritten.

Step ⇔ week alignment (seminar phases): Steps 1–2 ⇒ Weeks 1–7 (Setup + Evidence); Steps 3–4 ⇒ Weeks 4–11 (Design + Build, overlapping thin prototype); Step 5 ⇒ Weeks 8–11 (Calibrate); Steps 6–7 ⇒ Weeks 12–15 (Go-live + SP1/SP2 handoff). See Part 8 for milestone detail.

6.2.4 Scope knife — in / out

Use the scope knife. Park items for SP1/SP2 explicitly.

In scope (this seminar)	**Out of scope (park for SP1/SP2 / later)**
Role-aware login concept and dashboards for four roles (test accounts)	Replacing or API-integrating with the existing core internship system

Student status page and semester-wide status table with filters	Host-company self-service accounts
Host-company proposal submission, presentation scheduling, advisor decision, and decision history	Leave approval workflow (system records evidence after student handles leave with host company)
Readiness Gate logic showing four checks and blocking reasons	Production deployment or use as official decision-making record
Audit history for status changes and decisions in the implemented slice	Digital signatures and compensation/benefit processing
Synthetic test data, acceptance tests, scenario walkthroughs, documentation	Google Maps pins, route grouping, offline synchronization, letter generation, full completion workflow
Seminar paper documenting gap, method, results, limits	Mobile app, notifications, organization master, appeals, leave/evidence (Should-have)

6.2.5 Definition of the thin prototype this term

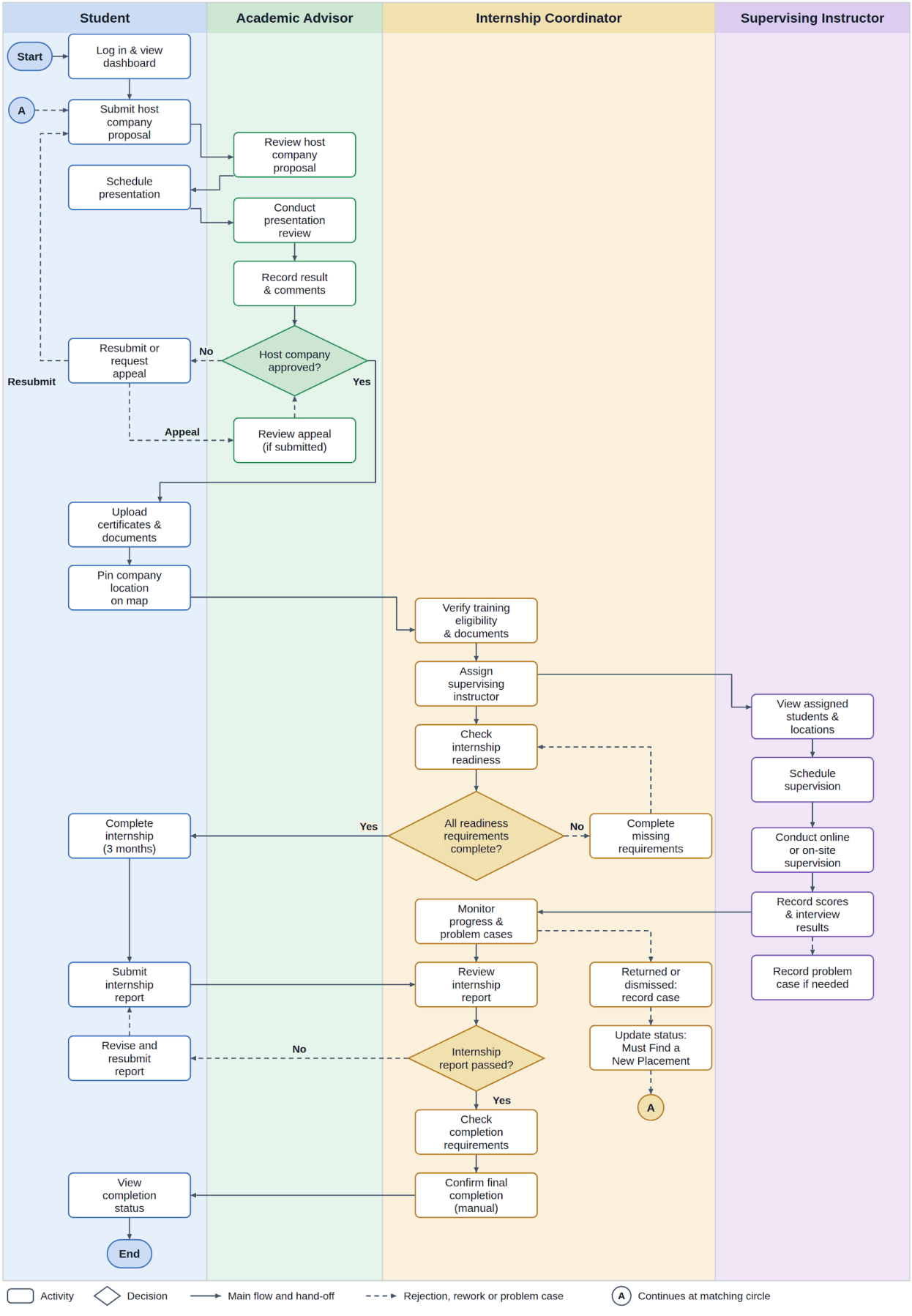
The thin prototype is an evaluable vertical slice, not a production-ready full system. It must demonstrate:

- A student's status and a semester-wide staff view.
- The host-company proposal and advisor decision loop.
- The four-condition Readiness Gate with visible blocking reasons.
- Auditable status/decision history.

It will use seeded synthetic records and test accounts. Maps, offline synchronization, letter generation, full completion workflow, and direct integration with the existing system may be represented in design artifacts and test plans but are not required to be fully implemented this term.

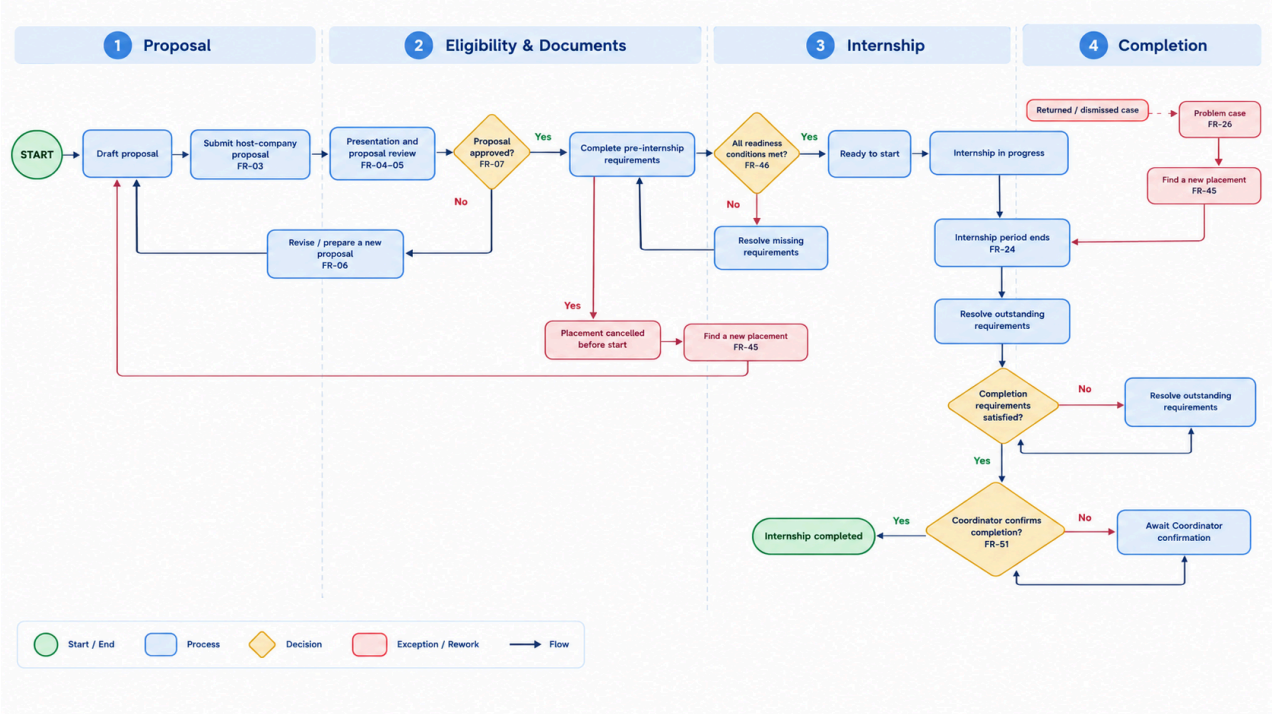
6.3 Role handoffs and swimlane workflow

The workflow below emphasizes handoffs rather than every screen. The Advisor owns the proposal presentation decision; the Coordinator verifies documents and training eligibility and manages instructor assignment; the Instructor carries out supervision; and the Coordinator confirms final closure after completion checks.

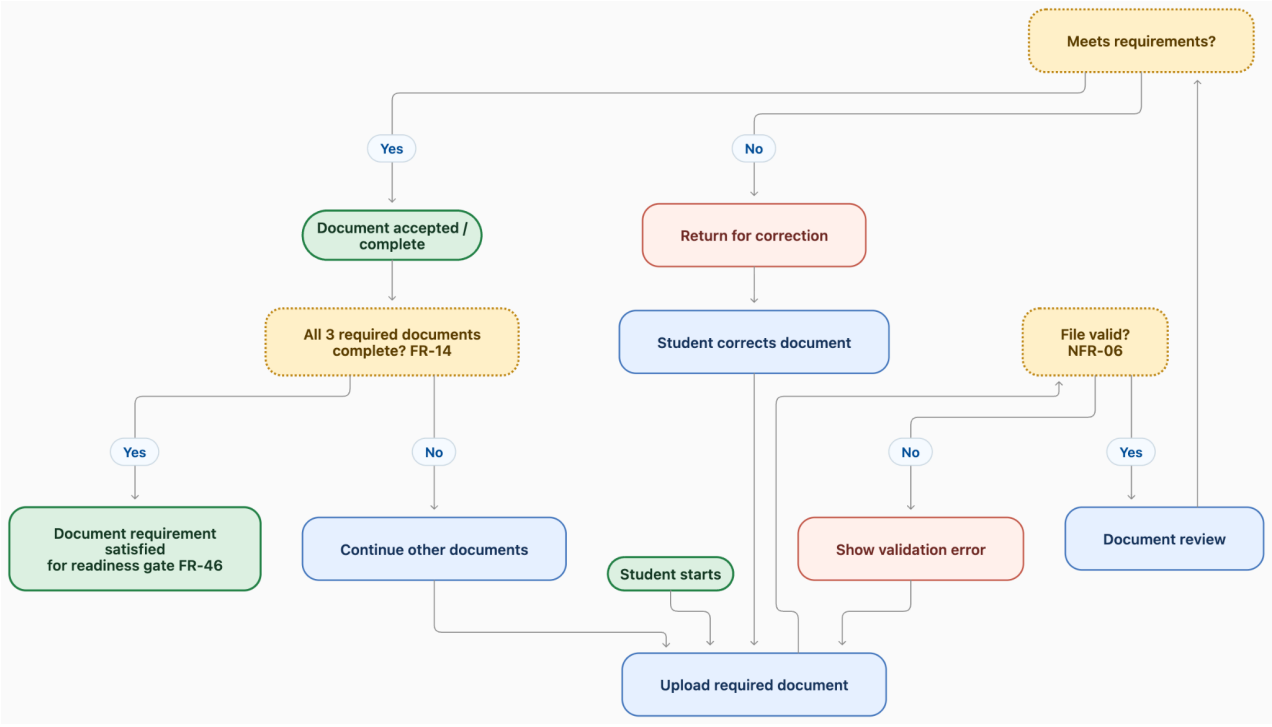

Student
Academic Advisor
Internship Coordinator
Supervising Instructor
Start
Log in & view dashboard
A
Submit host company proposal
Review host company proposal
Schedule presentation
Conduct presentation review
Record result & comments
Resubmit or request appeal
No
Host company approved?
Yes
Resubmit
Appeal
Review appeal (if submitted)
Upload certificates & documents
Pin company location on map
Verify training eligibility & documents
Assign supervising instructor
View assigned students & locations
Check internship readiness
Schedule supervision
Complete internship (3 months)
Yes
All readiness requirements complete?
No
Complete missing requirements
Conduct online or on-site supervision
Monitor progress & problem cases
Record scores & interview results
Submit internship report
Review internship report
Returned or dismissed: record case
Record problem case if needed
Revise and resubmit report
No
Internship report passed?
Update status: Must Find a New Placement
Yes
A
Check completion requirements
View completion status
Confirm final completion (manual)
End
Activity
Decision
Main flow and hand-off
Rejection, rework or problem case
A
Continues at matching circle

6.4 State Machine Diagram

6.4.1 Internship Whole State Changes



6.4.2 Document State Changes



6.5 Functional Requirements

Priorities follow the supplied feature list:

- Must have = required core capability
- Should have = planned capability with lower priority

6.5.1 Internship status tracking

Feature priority: Must have.

FR-01 [Must have]: The system shall show each student their own current internship status.

FR-02 [Must have]: The system shall provide Academic Advisors and Internship Coordinators with a semester-wide student status overview.

FR-03 [Must have]: The overview shall be filterable by status, including proposal pending, awaiting presentation, approved, missing documents, in internship, returned/dismissed, and internship ended.

FR-04 [Must have]: The system shall show readiness status as ready, not yet complete, or urgent when the start date is near and required conditions are incomplete.

6.5.2 Host-company proposal, approval, replacement, and appeal

Feature priority: Must have / Appeal Should have.

FR-05 [Must have]: The student shall be able to submit the desired host-company information.

FR-06 [Must have]: The student shall be able to schedule a proposal presentation with the Academic Advisor.

FR-07 [Must have]: The Academic Advisor shall record a passed/not-passed result and comments.

FR-08 [Must have]: If a proposal is not approved, the student shall be able to propose another company and schedule another presentation.

FR-09 [Appeal Should have]: The system shall retain previous proposal and decision history.

FR-10 [Appeal Should have]: If a host company cancels acceptance after approval but before the internship starts, the status shall change to “must find a new placement”; the original history shall remain linked.

FR-11 [Appeal Should have]: A student shall be able to request a review of a failed presentation result. The reason and second result shall be stored separately from the original result.

6.5.3 Training eligibility

Feature priority: Must have.

FR-12 [Must have]: The system shall track the required number of preparation-training sessions; the default is five and the value shall be configurable.

FR-13 [Must have]: The student shall be able to upload a certificate as evidence for each training session.

FR-14 [Must have]: The Internship Coordinator shall verify training attendance/certificates and record whether eligibility is complete.

FR-15 [Must have]: The student shall be able to request a review of a failed eligibility result; the request and second result shall be kept as separate history.

6.5.4 Internship documents and request letter

Feature priority: Must have.

FR-16 [Must have]: The system shall track the Request Letter, Acceptance Letter, and Referral/Placement Letter.

FR-17 [Must have]: The system shall show which required documents are present or missing.

FR-18 [Must have]: The system shall reject unsupported file types or files larger than 10 MB immediately and explain the reason.

FR-19 [Must have]: The system shall send reminders for missing documents and stepped follow-up reminders for an outstanding Acceptance Letter; reminder intervals shall be configurable.

6.5.5 Company locations, routes, and instructor assignment

Feature priority: Must have.

FR-20 [Must have]: The student shall be able to pin the host-company location on Google Maps.

FR-21 [Must have]: The system shall display pinned company locations to authorized staff.

FR-22 [Must have]: The system shall support grouping nearby companies to help plan supervision routes; the degree of automation is to be confirmed.

FR-23 [Must have]: The Internship Coordinator shall assign one Supervising Instructor per company/route group according to the defined assignment rule.

FR-24 [Must have]: If a company already has an assigned instructor, the system shall display the existing assignment and require confirmation before replacement.

FR-25 [Must have]: The system shall retain the history of instructor assignment changes, including old/new instructor, person making the change, and time.

6.5.6 Supervision scheduling, scoring, and interview records

Feature priority: Must have.

FR-26 [Must have]: The assigned Supervising Instructor shall schedule each supervision in advance with date, time, and format (online or on-site).

FR-27 [Must have]: The system shall show supervision scheduling status as not yet scheduled, scheduled, or supervision completed.

FR-28 [Must have]: The Supervising Instructor shall record student scores using the approved scoring form/rubric.

FR-29 [Must have]: The Supervising Instructor shall record the student interview result.

FR-30 [Must have]: For on-site supervision, the system shall save records as local drafts if connectivity is unavailable and synchronize them when connectivity returns.

FR-31 [Must have]: The system shall show a persistent pending-sync indication until offline data is synchronized successfully.

FR-32 [Must have]: The Supervising Instructor shall be able to record a problem case during supervision.

6.5.7 Existing-system confirmation

Feature priority: Must have.

FR-33 [Must have]: The student shall be able to declare that daily tasks/data have been recorded in the existing core internship system.

6.5.8 Internship end, evaluation, report, and closure

Feature priority: Must have.

FR-34 [Must have]: After the planned three-month duration is complete, the system shall change the status to internship ended.

FR-35 [Must have]: The system shall be able to generate a thank-you letter to the host company from a template using stored data.

FR-36 [Must have]: When the host-company evaluation is recorded as complete, the system shall notify the student.

FR-37 [Must have]: The student shall be able to submit an internship report.

FR-38 [Must have]: The system shall retain every report version when the report is revised and resubmitted.

FR-39 [Must have]: Authorized staff/Academic Advisor shall review the report, record comments, and set the result to passed or needs revision.

FR-40[Must have]: The system shall check the completion conditions: planned duration complete, required minimum supervision count complete, host-company evaluation complete, report passed, and leave records verified if any.

FR-41 [Must have]: The system shall display any unmet completion conditions and shall not enable final closure until all required conditions are met.

FR-42 [Must have]: The Internship Coordinator shall manually confirm final status closure; this action shall not be bypassed automatically.

6.5.9 Problem-case tracking

Feature priority: Must have.

FR-43 [Must have]: The Internship Coordinator or Supervising Instructor shall record returned/dismissed cases and students at risk of not passing.

FR-44 [Must have]: A problem-case record shall include date, reason, recorder, and assistance measures where applicable.

FR-45 [Must have]: The system shall show a chronological problem-case history for each student.

FR-46 [Must have]: When a student is returned/dismissed during the internship, the status shall change to must find a new placement and the student may restart the host-company proposal process.

6.5.10 Notifications and reminders

Feature priority: Should have.

FR-47 [Should have]: The system shall notify Academic Advisors when a proposal presentation is approaching.

FR-48 [Should have]: The system shall notify Supervising Instructors when a supervision schedule is approaching.

FR-49 [Should have]: Internship coordinator shall be able to configure reminder lead time. A seven-day default is suggested but is not confirmed.

6.5.11 Organization master and placement quota

Feature priority: Should have.

FR-50 [Should have]: When a student proposes a host company, the system shall search existing organization records first and allow selection of a matching record to reduce duplicates.

6.5.12 Readiness gate

Feature priority: Must have.

FR-51 [Must have]: Before the internship starts, the system shall check that the host-company proposal is approved, all three required documents are complete, training eligibility has passed, and a Supervising Instructor is assigned.

FR-52 [Must have]: The system shall prevent the internship status from changing to in internship while any required readiness condition is incomplete.

FR-53 [Must have]: The Internship Coordinator shall be able to filter the dashboard to show urgent readiness cases.

6.5.13 Leave and evidence records

Feature priority: Should have.

FR-54 [Should have]: The student shall be able to record leave type, date range, number of days, reason, and supporting evidence after handling the leave with the host company.

FR-55 [Should have]: The Supervising Instructor shall have read-only access to leave information and evidence.

FR-56 [Should have]: The student shall be able to upload general internship evidence such as activity photos or supporting documents.

FR-58 [Must have]: The system shall mark a readiness case as URGENT when the internship start date is within a configurable urgent window and at least one readiness condition is incomplete. The urgent window shall default to 14 calendar days and shall be configurable by the Internship Coordinator.

FR-59 [Must have]: The system shall display URGENT as a prioritization label only. It shall not change the readiness gate result, shall not allow the internship to start, and shall not bypass any required condition.

6.6 Main database tables and relationships

Table	Purpose
users	Login accounts and role-based access

students	Student internship profiles
host_companies	Internship companies and their locations
placement_proposals	Company placement proposals and presentation results
training_records	Pre-internship training and certificates
internship_documents	Required documents and verification
route_groups	Groups of nearby companies for supervision
instructor_assignments	Instructors assigned to route groups
supervisions	Scheduled and completed company visits
problem_cases	Internship problems and assistance actions
leave_records	Leave requests, evidence, and approval
internship_reports	Final reports and instructor reviews
notifications	System notifications
audit_history	Records of important system changes

6.7 Gate override policy

Proposed baseline: a Coordinator may not bypass a failed Readiness Gate or Completion Gate simply because paperwork is delayed. The system should show the missing condition, allow follow-up, and mark the case urgent when appropriate. This preserves the SRS business rule that students cannot start until all readiness conditions pass and final closure requires all completion conditions plus Coordinator confirmation.

If the university has a formal exception process, it must be approved by stakeholders before implementation. Any future override should require a distinct “exception approval” permission, a mandatory reason, reference to supporting evidence, timestamp, actor identity, affected conditions, and an immutable audit entry; it must never silently rewrite the gate result. A readiness override must not be enabled by default in this proposal, and final completion must not be automatically bypassed.

6.8 Proposed measurable non-functional targets

The SRS says exact performance and availability targets have not yet been supplied by stakeholders. The following are proposed prototype acceptance baselines—not confirmed university service-level commitments. Confirm them with the advisor and Coordinator and record any changes.

Quality area	Proposed baseline target	How to verify
Response time	For normal dashboard/status/search requests: 95% complete within 2 seconds under 25 concurrent test users using seeded data.	Load test; record p50/p95 and test environment.
Availability	At least 99% availability during a planned demo/test window, excluding announced maintenance.	Monitor test environment during agreed window; report downtime.
Concurrency	Support 25 simultaneous authenticated users in prototype load testing without failed workflow transactions.	Load test mixed read/write scenarios.
File validation	Accept PDF/JPG/PNG only, maximum 10 MB; reject invalid type/size with a clear message.	Automated positive and negative upload tests.
Audit integrity	100% of in-scope status/decision changes record actor, role, timestamp, old/new values, and reason where required.	Automated tests and database audit review.
Access control	100% of tested role-protected actions deny unauthorized roles.	Role-based authorization test matrix.
Offline draft safety	On-site draft remains visible and is not deleted before successful synchronization; pending-sync state persists.	Disconnect/reconnect test; SP2 full implementation target.
Recovery/retention	Backup schedule, recovery-time objective, recovery-point objective, and retention period to be agreed before production use.	Document approved policy; test restore before production.

The 2-second/25-user/99% values are intentionally modest prototype targets and should be reviewed against hosting capacity and course expectations. File upload duration and report generation should be measured separately if those functions are included in a test build.

6.7.1 Test environment baseline

All response-time, availability, and load measurements reported in this proposal will be recorded against a documented test environment so results are reproducible. The baseline environment for the seminar prototype is:

- **Hosting:** Local development machine for functional tests; a single small cloud instance (e.g., 1 vCPU, 2 GB RAM) for the load and availability smoke tests.
- **Database:** PostgreSQL 16, single instance, seeded with synthetic data.
- **Backend:** Spring Boot (Java 17) REST API, default JVM heap settings.
- **Frontend:** React production build served as static files.
- **Load generator:** k6 or Apache JMeter, run from the same local network as the test instance.
- **Seed data size:** Approximately 100 synthetic students, 50 host companies, and 300 status/audit records, so that dashboard and search queries reflect realistic table sizes.
- **Concurrency:** Up to 25 concurrent simulated users for the load test.
- **Measurement:** p50 and p95 response times recorded per endpoint; availability measured as successful responses divided by total requests over the agreed demo/test window.
- **Reporting:** The exact machine specifications, OS, Java version, PostgreSQL version, and test script version will be recorded with each test run in the test report.

If the environment changes (e.g., a different hosting tier is used for the defense demo), the change will be noted alongside the results, and the proposed targets in the table below will be re-checked against the new environment.

7. Evaluation and Success Criteria

7.1 Evaluation design

Evaluation combines engineering verification with reviewer feedback. The prototype will use synthetic student/company records and predefined scenarios, not real student documents. Recruit approximately 5–10 reviewers from the Coordinator, Advisor, Instructor, or software-engineering reviewer groups if access and time permit. Each reviewer completes 3–4 scenarios in approximately 20–30 minutes.

Metric	Operational definition	Target for seminar
Traceability	Each in-scope requirement has an acceptance criterion and at least one test case.	100% of requirements in the agreed seminar slice; full SRS traceability matrix drafted and gaps recorded.
Prototype correctness	Automated tests pass for implemented acceptance criteria.	100% of in-scope acceptance criteria pass before defense, or failures are explicitly reported.
Status clarity	The reviewer can identify the stage and blocker from the dashboard.	At least 70% of status-clarity ratings are 4 or 5 out of 5.
Auditability	Implemented status/decision changes contain required audit fields.	100% of tested changes audited; mean reviewer auditability score ≥4/5.
Gate correctness	Gate result matches the defined rule for seeded scenarios.	100% of defined readiness-gate test scenarios produce expected outcomes.
Assumption resolution	Open questions confirmed or parked with owner and due date.	Resolve or explicitly park at least 9 of the 11 open assumptions tracked in the proposal baseline; reconcile this count against the current SRS open-question list.

7.2 Example test scenarios

- A student has an approved proposal, passed training, an assigned instructor, but one required letter is missing: readiness must remain blocked.
- A presentation is rejected: the result and comments remain visible and the student can begin another proposal cycle.
- A host company cancels after approval but before the start: the status changes to must find a new placement and prior history remains linked.
- A student is near the planned start date with missing conditions: the dashboard marks the case urgent but does not allow start.
- A report needs revision: the earlier version remains available and the completion gate stays blocked.
- All completion conditions are met: the Coordinator must still perform the final manual confirmation.

7.3 Limitations

A small reviewer group and synthetic data support formative evaluation, not a claim of institution-wide effectiveness. If stakeholder recruitment fails, the team may use advisor-approved role-play scenarios and must state this limitation. No reduction in staff hours or improvement in student outcomes will be claimed without direct measurement.

8. Timeline, Seminar Defense, and SP1/SP2 Roadmap

The schedule below assumes a 15-week seminar term. Replace week labels with actual course dates once the end-of-term seminar and proposal-defense calendar is confirmed.

Period	Milestones and work	Evidence / deliverable
Weeks 1–3	Scope lock; review SRS and literature; ethics/privacy plan; stakeholder access request.	Proposal v2, question guide, risk and assumption registers.
Weeks 4–5	Stakeholder interviews; map current process; confirm roles, documents, and rules.	Interview summary; current-state workflow; evidence log.
Weeks 6–7	Freeze seminar slice; finalize state machine, gates, architecture, data model, NFR baselines.	Design review package; traceability matrix draft; prototype wireframes.
Weeks 8–9	Build dashboard/status table, proposal loop, readiness gate, and audit events.	Thin vertical-slice prototype with synthetic data.
Weeks 10–11	Automated tests, scenario walkthroughs, NFR smoke/load tests, feedback revisions.	Test results; reviewer scores; issue list.
Weeks 12–13	Analyze results; revise diagrams and requirements; draft report and defense presentation.	Results/limitations section; defense slides; demo script.
Weeks 14–15	Rehearse and deliver end-of-term seminar proposal defense; archive handoff package.	Final proposal/paper, working demo, SP1/SP2 roadmap and backlog.

8.1 Definition of the thin prototype this term

The thin prototype is an evaluable vertical slice, not a production-ready full system. It must demonstrate a student's status, a semester-wide staff view, the host-company proposal and advisor decision loop, the four-condition Readiness Gate, and auditable status/decision history. It will use seeded synthetic records and test accounts. Maps, offline synchronization, letter generation, full completion workflow, and direct integration with the existing system may be represented in design artifacts and test plans but are not required to be fully implemented this term.

8.2 Senior Project 1 roadmap

- Finalize requirements and resolve remaining stakeholder questions; update the complete traceability matrix.
- Implement core role-based workflows end to end: training eligibility, three documents, placement assignment, supervision scheduling and records, problem cases, and completion checklist.
- Implement database schema, authentication/authorization, audit history, file validation, and notifications according to confirmed scope.
- Conduct integration and usability testing with agreed synthetic or approved anonymized data.
- Deliver an integrated beta, test evidence, updated SRS, and documented unresolved risks.

8.3 Senior Project 2 roadmap

- Complete Google Maps company pins and route grouping; implement the agreed level of route-planning assistance.
- Implement offline supervision drafts, persistent pending-sync indication, retry behavior, and conflict/error handling.
- Complete internship-end actions, report versioning/review, evaluation notifications, and Completion Gate enforcement.
- Add remaining prioritized Should-have functions where feasible, such as reminders, appeals, leave/evidence, and organization master.
- Run broader acceptance, performance, privacy, and recovery tests; refine usability from stakeholder feedback.
- Prepare deployment, user guide, training, maintenance plan, and final project evaluation.

8.4 Defense readiness checklist

- Problem statement cites actual collected evidence and clearly labels remaining evidence gaps.
- Related work explains the specific gap rather than only listing systems.
- Demo shows the proposal loop, readiness blocker, and audit history.
- Swimlane, state machine, and architecture diagrams match the latest SRS.
- NFR targets are labeled, proposed or confirmed, with test methods.
- Gate override rule is explicitly approved or remains disabled.
- Budget, team effort, risks, and privacy safeguards are included.
- SP1 and SP2 work is separated from the seminar deliverable.

9. Team Effort, Budget, and Risk Management

9.1 Team roles and hours

The proposal estimates three team members over 15 weeks at approximately 8 hours per member per week, for 360 person-hours in total. The split below is a planning estimate and can be adjusted based on each member's strengths.

Workstream	Lead / support	Estimated team hours	Output
Requirements and evidence	Requirements lead; all members support interviews/review.	90	Evidence log, assumption register, traceability matrix.
Workflow and architecture	Design/prototype lead; team review.	75	Swimlane, state machine, architecture, data model.
Prototype implementation	Implementation lead; all members contribute.	120	Runnable thin slice.
Testing and evaluation	Testing/evaluation lead; team supports sessions.	45	Test suite, reviewer protocol, results.
Writing and defense	Shared by all members.	30	Proposal, slides, demo rehearsal.
Total	3 members × 15 weeks × 8 hours/week.	360	Seminar package.

9.2 Estimated cash budget

Item	Estimated cost (THB)	Basis / note
Hosting and database	0–800	Free tier preferred; low-cost hosting if needed for demo.
Development tools	0	Open-source tools and student-accessible tools where available.
Synthetic test data	0	Generated test records; no purchase required.
Reviewer sessions	0	In-kind participation; no payment currently planned.
Maps / file storage	0 during seminar prototype	Later use may incur charges; check quotas and terms before SP1/SP2 integration.
Total estimated seminar cash	0–800	Excludes team time and any later production cost.

9.3 Risk register

Risk	Likelihood / impact	Mitigation and trigger
Stakeholders unavailable	Medium / High	Request interviews early; if unavailable by Week 5, document

		attempts and use advisor-approved role-play scenarios.
Requirements continue changing	High / Medium	Version the SRS and change log; update acceptance criteria/tests when a rule changes; freeze seminar slice by Week 7.
Scope exceeds term capacity	High / High	Protect the thin slice; move maps, offline sync, letters, and full completion workflow to SP1/SP2 unless core slice is complete.
Gate rules are misunderstood	Medium / High	Review scenarios with Coordinator and Advisor; obtain written confirmation before implementation.
Privacy incident	Low / High	Use synthetic data by default, restrict access, avoid personal identifiers in notes, and remove any accidentally collected data promptly.
Offline sync causes duplicate/conflicting records	Medium / High	For later build, use unique draft IDs, server acknowledgments, retry-safe requests, and visible conflict handling; test before SP2 acceptance.
Map API cost or quota issue	Medium / Medium	Use limited test data and monitor quotas; keep map integration modular and provide a documented fallback for seminar demo.
NFR targets fail under test	Medium / Medium	Record environment and bottleneck; optimize or revise target with advisor approval rather than claiming success without evidence.

10. Data Privacy, PDPA, and Governance

10.1 Data minimization

The seminar prototype will use synthetic student, company, document, and status records. Real student identifiers, phone numbers, certificates, letters, reports, leave evidence, and other personal data will not be copied into the prototype. If stakeholder interviews are conducted, notes should use role labels or participant codes rather than names unless identification is necessary and explicitly agreed.

10.2 Consent and purpose

- Explain the project purpose, voluntary participation, expected duration, how feedback will be used, and that the prototype does not make official internship decisions.
- Obtain consent before recording interview notes or using direct quotations; anonymize quotations in the proposal unless permission is explicit.
- Use feedback only for requirements validation and academic evaluation, not for assessing individual students or staff.

10.3 Access control and security

- Require authentication for protected pages and enforce role permissions on the backend, not only in the React interface.
- Limit access to project data to named team members and the faculty advisor; use test accounts for demos.
- Store passwords as hashes, protect secrets/API keys, and avoid committing credentials or personal data to source control.
- Validate file type and size; restrict access to uploaded files and use secure storage configuration when file upload is implemented.
- Log important decisions and changes, but avoid putting unnecessary sensitive content into audit logs.

10.4 PDPA handling plan

Because the project may eventually handle identifiable student and staff information, the team will apply privacy-by-design principles and confirm the applicable university procedures and Thailand's Personal Data Protection Act (PDPA) requirements with the advisor or responsible university office before collecting or processing real personal data. This proposal is a project plan, not a legal determination of the university's role or lawful basis.

Topic	Proposed handling
Purpose and lawful basis	Document why each data field is needed and confirm the appropriate lawful basis/notice with the university before real-data use.
Collection notice	Explain what is collected, purpose, access, retention, contact point, and participant rights using university-approved wording.
Data minimization	Collect only required fields; use synthetic data for development and demonstrations.

Storage and transfer	Use access-controlled storage; avoid personal data in public repositories, personal drives, or unapproved third-party services.
Retention and deletion	Agree retention periods with the university; delete interview notes and test artifacts when no longer needed, subject to course requirements and approved policy.
Data-subject requests / incidents	Escalate correction, access, deletion, or suspected incident requests to the advisor and university's designated contact; do not handle them informally.
Third-party services	Do not send personal data to Google Maps or other third parties. Map requests should use company location data only where approved and necessary.

10.4.1 Retention and deletion for the seminar prototype

Because the seminar prototype uses only synthetic student, company, document, and status records generated by the team, no retention or deletion policy for real personal data is required during the seminar term. Synthetic records contain no personal data as defined under Thailand's PDPA and may be retained, regenerated, or deleted freely by the team for development, testing, and demonstration purposes.

The following rules apply specifically to the seminar prototype:

- Synthetic data only. No real student identifiers, phone numbers, certificates, letters, reports, leave evidence, or other personal data will be stored in the prototype database, file storage, or backups.
- No production retention obligation. Because no real personal data is processed, the prototype is not subject to the university's records-retention schedule during the seminar term.
- Deletion on demand. Any synthetic dataset used for a demo may be dropped and regenerated at any time without notice, since it contains no personal data.
- Interview notes. If stakeholder interviews are conducted, notes will use role labels or participant codes rather than names, and will be retained only until the end of the seminar term, after which they will be deleted by the team.
- Trigger for a full retention policy. A formal retention, deletion, and subject-rights policy will be drafted and reviewed with the advisor and the responsible university office before any real personal data is collected or processed (i.e., before SP1 if real data is used, or before any pilot deployment). This is tracked as OQ-10 in Appendix D.

10.5 Ethics statement

The prototype is for academic evaluation and will not replace official decisions by the Academic Advisor or Internship Coordinator. Gate outcomes and audit history must be transparent. Any use of real data, external hosting, or third-party service must be reviewed and approved before it occurs.

Appendix A. Proposed Gate Conditions and Override Audit

A1. Readiness Gate

- Host-company proposal is approved by the Academic Advisor.
- Request Letter, Acceptance Letter, and Referral/Placement Letter are all present and verified as required.
- Training eligibility is passed according to the configured session requirement and certificate checks.
- A Supervising Instructor is assigned to the company/route group.

Outcome: Ready only when all four conditions pass. Otherwise, Not Yet Complete; Urgent may be displayed when the start date is near. Urgent is a prioritization label, not a pass.

A2. Completion Gate

- The planned internship duration is complete.
- Minimum supervision count is satisfied according to the confirmed rule.
- Host-company evaluation is recorded as complete.
- Internship report is passed by authorized reviewer; report revisions preserve earlier versions.
- Leave records are verified if any exist.
- The coordinator manually confirms final closure.

A3. Audit fields for decisions and exceptions

Field	Purpose
event_id	Unique audit event.
actor_id / role	Who performed the action and under which role.
timestamp	When the action occurred.
action and reason	What happened and why; reason required for any approved exception.
old_value / new_value	Previous and new status or decision.
gate conditions snapshot	Values of each required condition at the time of evaluation.
exception reference	If a formal exception process is later approved, link to the approver and supporting evidence.
result	Approved, rejected, blocked, or reverted; previous events remain available.

Appendix B. Reviewer Evaluation Rubric

Use a 1–5 scale (1 = strongly disagree; 5 = strongly agree) after each scenario.

ID	Statement
E1 Status clarity	I can identify the student's current stage and what blocks the next step.
E2 Gate correctness	The readiness result matches the defined rule and scenario facts.
E3 Auditability	I can see who changed a status, when, and what changed.
E4 Role fit	Each role sees only the information and actions appropriate to it.
E5 Workflow fit	Proposal rejection, replacement placement, and case handling match the agreed process.
E6 Actionability	I can identify which student or case needs follow-up first.

Success criteria: at least 70% of status-clarity ratings are 4/5 or higher; mean auditability score is at least 4/5; all in-scope acceptance tests pass or failures are transparently documented; all tested state changes are audited.

Appendix C. References

[1] M. Z. Husin, P. R. Ganes, and N. Puteh, "A Web-Based Industrial Training Management System: A Case Study of the School of Computing," Journal of Digital System Development, vol. 2, no. 2, pp. 64–78, 2024, doi: 10.32890/jdsd2024.2.2.5.

[2] H. M. C. Pushpakumara, "Students Internship Management System," M.Sc. dissertation, University of Colombo School of Computing, 2013. http://hdl.handle.net/123456789/594

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[4] B. Moses, "Internship Management Information System Case Study: Soroti University, Soroti," Research Report, Busitema University. https://bdears.busitema.ac.ug/handle/123456789/2155

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[7] C. Radl, A Systematic Literature Review on Requirements Tracing Approaches, M.Sc. thesis, Technische Universität Wien, 2007.

[8] Internship Navigator, Software Requirements Specification, Version 1.0, 9 October 2026 (project document).

[9] Project documentation set, “se-internship-system” (requirements specification, adopted functions, backlog, feature list, user journeys, test plan, acceptance criteria).
https://github.com/prascymj/se-internship-system

Appendix D. Open Questions for Stakeholder Confirmation

ID	Question to confirm	Owner / target
OQ-01	Who records the host-company evaluation: Coordinator/Advisor on behalf of the company, or a future company account?	Coordinator / Week 5
OQ-02	What reminder lead times and notification channels should be used?	Coordinator / Week 6
OQ-03	Should route grouping be manual, assisted, or automatically optimized?	Coordinator / SP1 design
OQ-04	Confirm training-session count, minimum supervision count, cut-off dates, holidays, and official templates.	Coordinator / Week 6
OQ-05	Which file-storage provider and access/retention rules are approved?	Advisor / before real uploads
OQ-06	Who creates accounts, resets passwords, and maintains role assignments?	System owner / SP1
OQ-07	Should Request Letters be generated or uploaded manually?	Coordinator / Week 6
OQ-08	Approve or revise proposed performance, availability, concurrency, backup, recovery, and retention targets.	Advisor + Coordinator / Week 7
OQ-09	Does the university have a formal gate exception process? If yes, who may authorize it and what evidence is required?	Advisor + Coordinator / before implementation
OQ-10	What university-approved privacy notice, consent, retention, and incident-escalation process applies?	Advisor / before research with real data